

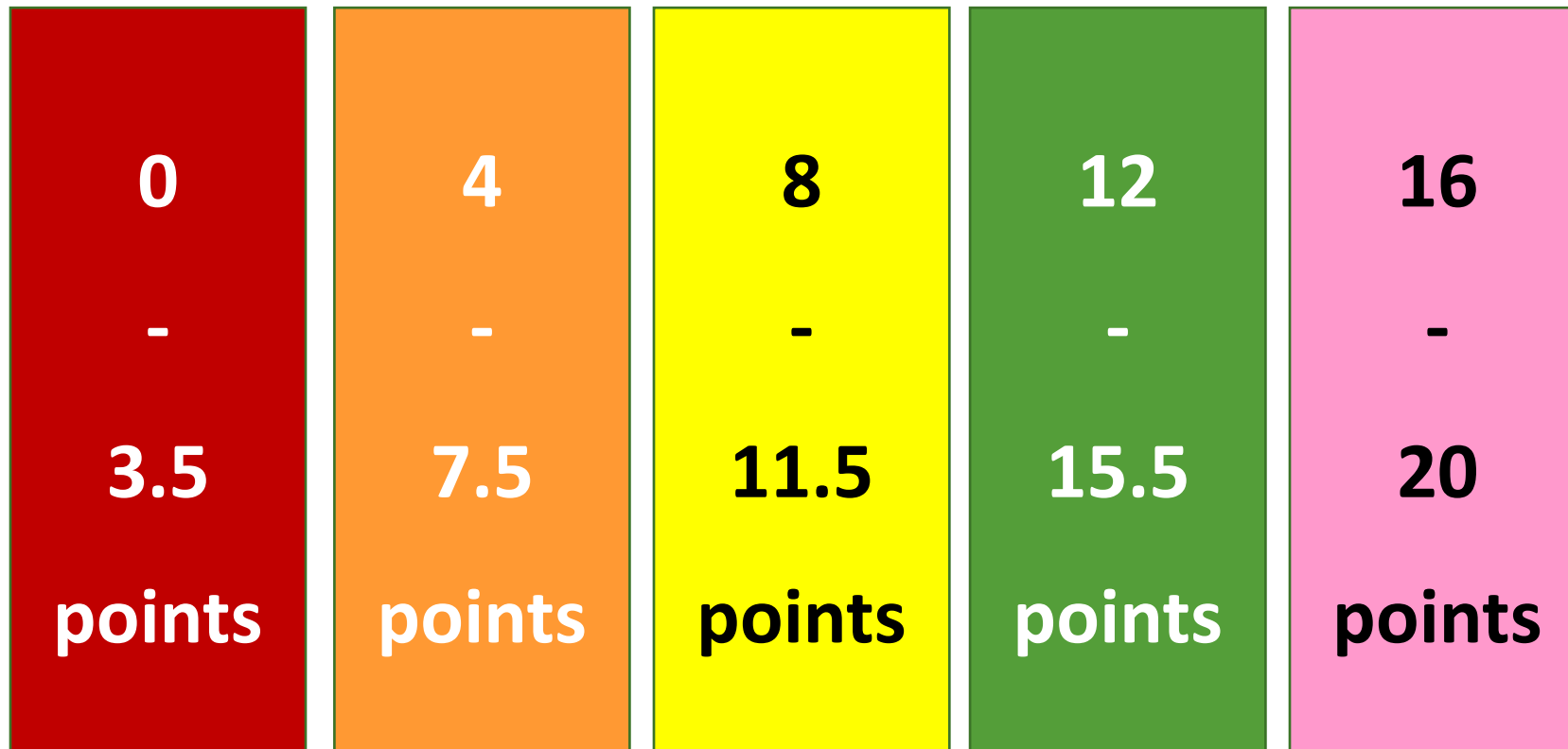
Lecture 12

Revision 01 - Inner Products

The Wooclap scores are available on NTULearn.

Which box do you belong to?

To obtain one point, you need to answer TRUTHFULLY.



QUESTION 3.**(20 marks)**

Consider the vector space $\mathcal{P}_3(\mathbb{R})$ and the following three polynomials:

$$q_0(x) = \frac{(x-1)(x-2)(x-3)}{-6},$$

$$q_1(x) = \frac{x(x-2)(x-3)}{2},$$

$$q_2(x) = \frac{x(x-1)(x-3)}{-2}.$$

Throughout this question, we consider the following product function

$$\langle f(x), g(x) \rangle \triangleq f(0)g(0) + f(1)g(1) + f(2)g(2) + f(3)g(3).$$

Also, throughout this question, you may use the following fact: if two polynomials $f(x)$ and $g(x)$ in $\mathcal{P}_3(\mathbb{R})$ evaluate to the same values on four distinct points, then $f(x)$ and $g(x)$ are the same polynomial.

- (a) Verify that the product function is a real inner product on $\mathcal{P}_3(\mathbb{R})$.
- (b) Show that $\{q_0(x), q_1(x), q_2(x)\}$ is *orthonormal* with respect to the inner product.
- (c) Let $f(x) = x^3$. By evaluating $f(x)$, $q_0(x)$, $q_1(x)$, and $q_2(x)$ at $x = 3$, explain why $f(x)$ does not belong to the span of $\{q_0(x), q_1(x), q_2(x)\}$. Then explain why $\{q_0(x), q_1(x), q_2(x), f(x)\}$ is a basis for $\mathcal{P}_3(\mathbb{R})$.
- (d) Apply the Gram-Schmidt procedure on the set $\{q_0(x), q_1(x), q_2(x), f(x)\}$ to find $q_3(x)$ such that $\mathcal{B} = \{q_0(x), q_1(x), q_2(x), q_3(x)\}$ is an orthonormal basis for $\mathcal{P}_3(\mathbb{R})$. Verify that

$$q_3(x) = \frac{x(x-1)(x-2)}{6}.$$

- (e) For a positive integer N , define $S(N) \triangleq 0^2 + 1^2 + \dots + N^2$. In other words, $S(N)$ is the sum of the first N squares. Last night, you dreamed that you derived a “nice” formula for $S(N)$. However, when you woke up in the morning, you forgot the formula. Nevertheless, you clearly remember that $S(N)$ is a polynomial in N and has degree exactly three. Write $S(N)$ as a linear combination of polynomials in $\mathcal{B}$.

Throughout this question, we consider the following product function

$$\langle f(x), g(x) \rangle \triangleq f(0)g(0) + f(1)g(1) + f(2)g(2) + f(3)g(3).$$

Also, throughout this question, you may use the following fact: if two polynomials $f(x)$ and $g(x)$ in $\mathcal{P}_3(\mathbb{R})$ evaluate to the same values on four distinct points, then $f(x)$ and $g(x)$ are the same polynomial.

(a) Verify that the product function is a real inner product on $\mathcal{P}_3(\mathbb{R})$.

What are the five properties that you need to verify?

This is NOT a trick question.

Just check off the five properties listed in class.

- | | |
|--|---|
| 1. Positivity | 5. Homogeneity in First Slot |
| 2. Definiteness | 6. Homogeneity in Second Slot |
| 3. Additivity in First Slot | 7. Symmetry |
| 4. Additivity in Second Slot | 8. Conjugate Symmetry |

Throughout this question, we consider the following product function

$$\langle f(x), g(x) \rangle \triangleq f(0)g(0) + f(1)g(1) + f(2)g(2) + f(3)g(3).$$

Also, throughout this question, you may use the following fact: if two polynomials $f(x)$ and $g(x)$ in $\mathcal{P}_3(\mathbb{R})$ evaluate to the same values on four distinct points, then $f(x)$ and $g(x)$ are the same polynomial.

(a) Verify that the product function is a real inner product on $\mathcal{P}_3(\mathbb{R})$.

Let's verify that
it is definite

$$\langle v, v \rangle = 0 \text{ iff } v = 0$$

($\Leftarrow$) Consider $z(x)$

$$\langle z(x), z(x) \rangle = z(0)^2 + z(1)^2 + z(2)^2 + z(3)^2 = 0$$

$$\Rightarrow \langle f(x), f(x) \rangle = f(0)^2 + f(1)^2 + f(2)^2 + f(3)^2 = 0$$

$$\Rightarrow f(0) = f(1) = f(2) = f(3) = 0 \text{ because } f(0), f(1), f(2), f(3) \text{ are real.}$$

$$\Rightarrow f(x) \text{ is zero poly}$$

Throughout this question, we consider the following product function

$$\langle f(x), g(x) \rangle \triangleq f(0)g(0) + f(1)g(1) + f(2)g(2) + f(3)g(3).$$

Also, throughout this question, you may use the following fact: if two polynomials $f(x)$ and $g(x)$ in $\mathcal{P}_3(\mathbb{R})$ evaluate to the same values on four distinct points, then $f(x)$ and $g(x)$ are the same polynomial.

(a) Verify that the product function is a real inner product on $\mathcal{P}_3(\mathbb{R})$.

Can you verify the remaining properties?

$$q_0(x) = \frac{(x-1)(x-2)(x-3)}{-6}, \quad \langle f(x), g(x) \rangle \triangleq f(0)g(0) + f(1)g(1) + f(2)g(2) + f(3)g(3).$$

$$q_1(x) = \frac{x(x-2)(x-3)}{2},$$

$$q_2(x) = \frac{x(x-1)(x-3)}{-2}.$$

(b) Show that $\{q_0(x), q_1(x), q_2(x)\}$ is *orthonormal* with respect to the inner product.

To show that the set is orthonormal, what do you need to show?

- ~~1.~~ $\langle q_i(x), q_j(x) \rangle = 0$ whenever $i \neq j$
2. $\langle q_i(x), q_j(x) \rangle = 1$ whenever $i \neq j$ ✗
3. $\langle q_i(x), q_i(x) \rangle = 0$ for $i = 0, 1, 2$ ✗
- ~~4.~~ $\langle q_i(x), q_i(x) \rangle = 1$ for $i = 0, 1, 2$

$$q_0(x) = \frac{(x-1)(x-2)(x-3)}{-6},$$

$$q_1(x) = \frac{x(x-2)(x-3)}{2},$$

$$q_2(x) = \frac{x(x-1)(x-3)}{-2}.$$

$$\langle f(x), g(x) \rangle \triangleq f(0)g(0) + f(1)g(1) + f(2)g(2) + f(3)g(3).$$

(b) Show that $\{q_0(x), q_1(x), q_2(x)\}$ is *orthonormal* with respect to the inner product.

Let us compute...

$$\begin{aligned} & \langle q_0(x), q_0(x) \rangle \\ &= q_0(0)q_0(0) + q_0(1)q_0(1) + \\ & \quad q_0(2)q_0(2) + q_0(3)q_0(3) \\ &= 1 \end{aligned}$$

$$\begin{aligned} & \langle q_1(x), q_0(x) \rangle \\ &= q_1(0)q_0(0) + q_1(1)q_0(1) \\ & \quad + q_1(2)q_0(2) + q_1(3)q_0(3) \\ &= 0 \end{aligned}$$

$$\begin{aligned} & \langle q_2(x), q_0(x) \rangle \\ &= q_2(0)q_0(0) + q_2(1)q_0(1) \\ & \quad + q_2(2)q_0(2) + q_2(3)q_0(3) \\ &= 0 \end{aligned}$$

$$q_0(x) = \frac{(x-1)(x-2)(x-3)}{-6},$$

$$q_1(x) = \frac{x(x-2)(x-3)}{2},$$

$$q_2(x) = \frac{x(x-1)(x-3)}{-2}.$$

$$\langle f(x), g(x) \rangle \triangleq f(0)g(0) + f(1)g(1) + f(2)g(2) + f(3)g(3).$$

(b) Show that $\{q_0(x), q_1(x), q_2(x)\}$ is *orthonormal* with respect to the inner product.

Let us compute...

$$\langle q_0(x), q_1(x) \rangle$$

$$\langle q_1(x), q_1(x) \rangle$$

$$\langle q_2(x), q_1(x) \rangle$$

$$q_0(x) = \frac{(x-1)(x-2)(x-3)}{-6}, \quad \langle f(x), g(x) \rangle \triangleq f(0)g(0) + f(1)g(1) + f(2)g(2) + f(3)g(3).$$

$$q_1(x) = \frac{x(x-2)(x-3)}{2},$$

$$q_2(x) = \frac{x(x-1)(x-3)}{-2}.$$

(b) Show that $\{q_0(x), q_1(x), q_2(x)\}$ is *orthonormal* with respect to the inner product.

Let us compute...

$$\langle q_0(x), q_2(x) \rangle$$

$$\langle q_1(x), q_2(x) \rangle$$

$$\langle q_2(x), q_2(x) \rangle$$

$$q_0(x) = \frac{(x-1)(x-2)(x-3)}{-6}, \quad \langle f(x), g(x) \rangle \triangleq f(0)g(0) + f(1)g(1) + f(2)g(2) + f(3)g(3).$$

$$q_1(x) = \frac{x(x-2)(x-3)}{2},$$

$$q_2(x) = \frac{x(x-1)(x-3)}{-2}.$$

(b) Show that $\{q_0(x), q_1(x), q_2(x)\}$ is *orthonormal* with respect to the inner product.

After these computations, can you quickly compute $\langle f(x), q_i(x) \rangle$ for any polynomial f ?

Either answer will reward one mark.

1. Yes
2. No

$$q_0(x) = \frac{(x-1)(x-2)(x-3)}{-6},$$

$$q_1(x) = \frac{x(x-2)(x-3)}{2},$$

$$q_2(x) = \frac{x(x-1)(x-3)}{-2}.$$

$$\langle f(x), g(x) \rangle \triangleq f(0)g(0) + f(1)g(1) + f(2)g(2) + f(3)g(3).$$

(c) Let $f(x) = x^3$. By evaluating $f(x)$, $q_0(x)$, $q_1(x)$, and $q_2(x)$ at $x = 3$, explain why $f(x)$ does not belong to the span of $\{q_0(x), q_1(x), q_2(x)\}$. Then explain why $\{q_0(x), q_1(x), q_2(x), f(x)\}$ is a basis for $\mathcal{P}_3(\mathbb{R})$.

$f(x)$ belongs to the span of $\{q_0(x), q_1(x), q_2(x)\}$ means

$$f(x) = \lambda_0 q_0(x) + \lambda_1 q_1(x) + \lambda_2 q_2(x) \dots$$

1. For some real numbers $\lambda_0, \lambda_1, \lambda_2$.
2. For some real polynomials $\lambda_0, \lambda_1, \lambda_2$.

$$q_0(x) = \frac{(x-1)(x-2)(x-3)}{-6},$$

$$\langle f(x), g(x) \rangle \triangleq f(0)g(0) + f(1)g(1) + f(2)g(2) + f(3)g(3).$$

$$q_1(x) = \frac{x(x-2)(x-3)}{2},$$

$$q_2(x) = \frac{x(x-1)(x-3)}{-2}.$$

(c) Let $f(x) = x^3$. By evaluating $f(x)$, $q_0(x)$, $q_1(x)$, and $q_2(x)$ at $x = 3$, explain why $f(x)$ does not belong to the span of $\{q_0(x), q_1(x), q_2(x)\}$. Then explain why $\{q_0(x), q_1(x), q_2(x), f(x)\}$ is a basis for $\mathcal{P}_3(\mathbb{R})$.

Prove by
contradiction

$f(x)$ belongs to the span of $\{q_0(x), q_1(x), q_2(x)\}$ means

$$f(x) = \lambda_0 q_0(x) + \lambda_1 q_1(x) + \lambda_2 q_2(x) \dots \text{--- Eq (1)}$$

1. For some real numbers $\lambda_0, \lambda_1, \lambda_2$.

At $x=3$, $\left. \begin{array}{l} \text{RHS} = 0 \\ \text{LHS} = 27 \end{array} \right\} \text{LHS} \neq \text{RHS, contradiction}$

$\therefore f(x)$ does NOT belong to span $\{q_0, q_1, q_2\}$

$$q_0(x) = \frac{(x-1)(x-2)(x-3)}{-6},$$

$$\langle f(x), g(x) \rangle \triangleq f(0)g(0) + f(1)g(1) + f(2)g(2) + f(3)g(3).$$

$$q_1(x) = \frac{x(x-2)(x-3)}{2},$$

$$q_2(x) = \frac{x(x-1)(x-3)}{-2}.$$

(c) Let $f(x) = x^3$. By evaluating $f(x)$, $q_0(x)$, $q_1(x)$, and $q_2(x)$ at $x = 3$, explain why $f(x)$ does not belong to the span of $\{q_0(x), q_1(x), q_2(x)\}$. Then explain why $\{q_0(x), q_1(x), q_2(x), f(x)\}$ is a basis for $\mathcal{P}_3(\mathbb{R})$.

$f(x)$ does not belong to the span of $\{q_0(x), q_1(x), q_2(x)\}$.

Claim: $\{q_0, q_1, q_2, f\}$ is L.I.

Suppose otherwise. There is a nontrivial solⁿ to

$$\lambda_0 q_0(x) + \lambda_1 q_1(x) + \lambda_2 q_2(x) + \lambda_3 f(x) = 0 \quad (*)$$

Case 1:

$\lambda_3 \neq 0$ Then $f(x)$ belongs to span $\{q_0, q_1, q_2\} \Rightarrow$ contradiction.

Case 2: $\lambda_3 = 0$. $\{q_0, q_1, q_2\}$ is NOT L.I. $\Rightarrow \{q_0, q_1, q_2\}$ is NOT orthogonal contradiction.

$$q_0(x) = \frac{(x-1)(x-2)(x-3)}{-6},$$

$$\langle f(x), g(x) \rangle \triangleq f(0)g(0) + f(1)g(1) + f(2)g(2) + f(3)g(3).$$

$$q_1(x) = \frac{x(x-2)(x-3)}{2},$$

$$q_2(x) = \frac{x(x-1)(x-3)}{-2}.$$

(c) Let $f(x) = x^3$. By evaluating $f(x)$, $q_0(x)$, $q_1(x)$, and $q_2(x)$ at $x = 3$, explain why $f(x)$ does not belong to the span of $\{q_0(x), q_1(x), q_2(x)\}$. Then explain why $\{q_0(x), q_1(x), q_2(x), f(x)\}$ is a basis for $\mathcal{P}_3(\mathbb{R})$.

$\{q_0(x), q_1(x), q_2(x), f(x)\}$ is linearly independent.

size of set is four which equals $\dim(\mathcal{P}_3(\mathbb{R})) = 4$

$\therefore \{q_0, q_1, q_2, f\}$ is a basis for $\mathcal{P}_3(\mathbb{R})$.

Ⓢ There are many other arguments

- (d) Apply the Gram-Schmidt procedure on the set $\{q_0(x), q_1(x), q_2(x), f(x)\}$ to find $q_3(x)$ such that $\mathcal{B} = \{q_0(x), q_1(x), q_2(x), q_3(x)\}$ is an orthonormal basis for $\mathcal{P}_3(\mathbb{R})$. Verify that

$$q_3(x) = \frac{x(x-1)(x-2)}{6}.$$

$\overset{u_1}{q_0}, \overset{u_2}{q_1}, \overset{u_3}{q_2}, \overset{u_4}{f}$ basis

$\downarrow$
 $\overset{v_1}{q_0}, \overset{v_2}{q_1}, \overset{v_3}{q_2}, \overset{v_4}{?}$ orthonormal basis

$\downarrow$ Because $[q_0, q_1, q_2]$ is orthogonal,

we can assume

$$v_1 = q_0, \quad v_2 = q_1, \quad v_3 = q_2$$

(d) Apply the Gram-Schmidt procedure on the set $\{q_0(x), q_1(x), q_2(x), f(x)\}$ to find $q_3(x)$ such that $\mathcal{B} = \{q_0(x), q_1(x), q_2(x), q_3(x)\}$ is an orthonormal basis for $\mathcal{P}_3(\mathbb{R})$. Verify that

$$q_3(x) = \frac{x(x-1)(x-2)}{6}.$$

wooclap

Suppose we are applying Gram-Schmidt procedure.

Say $\mathbf{u}_4 = f(x) = x^3$ and $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \{q_0(x), q_1(x), q_2(x)\}$.

Then

$$\mathbf{v}_4 = A - \frac{\langle A, B \rangle}{\langle B, B \rangle} q_0(x) - \frac{\langle A, C \rangle}{\langle C, C \rangle} q_1(x) - \frac{\langle A, D \rangle}{\langle D, D \rangle} q_2(x).$$

Which of the following is/are correct?

1. $A = x^2$

~~2.~~ $A = x^3$

~~3.~~ $B = q_0(x), C = q_1(x), D = q_2(x).$

4. $B = q_1(x), C = q_2(x), D = q_3(x).$

(d) Apply the Gram-Schmidt procedure on the set $\{q_0(x), q_1(x), q_2(x), f(x)\}$ to find $q_3(x)$ such that $\mathcal{B} = \{q_0(x), q_1(x), q_2(x), q_3(x)\}$ is an orthonormal basis for $\mathcal{P}_3(\mathbb{R})$. Verify that

$\langle q_0, q_0 \rangle = 1 \Rightarrow$ we don't need divide again.
 $q_3(x) = \frac{x(x-1)(x-2)}{6}.$

$$v_4 = f(x) - \langle f(x), q_0 \rangle q_0 - \langle f(x), q_1 \rangle q_1 - \langle f(x), q_2 \rangle q_2$$

$$\begin{aligned} \langle f(x), q_0 \rangle &= f(0)q_0(0) + f(1)q_0(1) + f(2)q_0(2) + f(3)q_0(3) \\ &= f(0) = 0 \end{aligned}$$

HIDDEN

- (d) Apply the Gram-Schmidt procedure on the set $\{q_0(x), q_1(x), q_2(x), f(x)\}$ to find $q_3(x)$ such that $\mathcal{B} = \{q_0(x), q_1(x), q_2(x), q_3(x)\}$ is an orthonormal basis for $\mathcal{P}_3(\mathbb{R})$. Verify that

$$q_3(x) = \frac{x(x-1)(x-2)}{6}.$$

HIDDEN

$$v_4 = x^3 - 0 \cdot q_0(x) - 1 \cdot q_1(x) - 8 \cdot q_2(x)$$

$$\begin{aligned} \langle v_4, v_4 \rangle &= v_4(0)^2 + v_4(1)^2 + v_4(2)^2 + v_4(3)^2 \\ &= 0^2 + 0^2 + 0^2 + 27^2 \end{aligned}$$

$$\text{So } \|v_4\| = 27$$

$$q_3(x) = \frac{v_4}{\|v_4\|} = \frac{x^3 - 0 \cdot q_0 - 1 \cdot q_1 - 8 \cdot q_2}{27}$$

(d) Apply the Gram-Schmidt procedure on the set $\{q_0(x), q_1(x), q_2(x), f(x)\}$ to find $q_3(x)$ such that $\mathcal{B} = \{q_0(x), q_1(x), q_2(x), q_3(x)\}$ is an orthonormal basis for $\mathcal{P}_3(\mathbb{R})$. Verify that

$$q_3(x) = \frac{x(x-1)(x-2)}{6}.$$

HIDDEN

$$q_3(x) = \frac{x^3 - 0 \cdot q_0(x) - 1 \cdot q_1(x) - 8 \cdot q_2(x)}{27} = \frac{x(x-1)(x-2)}{6}$$

① There is a trick to show " $=$ " without expanding.

Check that LHS = RHS at $x = 0, 1, 2, 3$

and we yellow part

→ remains to show

$$q_0(x) = \frac{(x-1)(x-2)(x-3)}{-6}, \quad \langle f(x), g(x) \rangle \triangleq f(0)g(0) + f(1)g(1) + f(2)g(2) + f(3)g(3).$$

$$q_1(x) = \frac{x(x-2)(x-3)}{2},$$

$$q_2(x) = \frac{x(x-1)(x-3)}{-2}, \quad q_3(x) = \frac{x(x-1)(x-2)}{6}.$$

After these computations, can you quickly compute $\langle f(x), q_i(x) \rangle$ for any polynomial f ?
 Either answer will reward one mark.

1. Yes
2. No

(e) For a positive integer N , define $S(N) \triangleq 0^2 + 1^2 + \dots + N^2$. In other words, $S(N)$ is the sum of the first N squares. Last night, you dreamed that you derived a “nice” formula for $S(N)$. However, when you woke up in the morning, you forgot the formula. Nevertheless, you clearly remember that $S(N)$ is a polynomial in N and has degree exactly three. Write $S(N)$ as a linear combination of polynomials in $\mathcal{B} = [q_0, q_1, q_2, q_3]$

$$\begin{aligned} S(N) &\triangleq 0^2 + 1^2 + \dots + N^2 = a_0 + a_1 N + a_2 N^2 + a_3 N^3 \\ &= \lambda_0 q_0(N) + \lambda_1 q_1(N) + \lambda_2 q_2(N) + \lambda_3 q_3(N) \end{aligned}$$

What is our task?

1. Find a_0, a_1, a_2, a_3

✓ 2. Find $\lambda_0, \lambda_1, \lambda_2, \lambda_3$

(e) For a positive integer N , define $S(N) \triangleq 0^2 + 1^2 + \dots + N^2$. In other words, $S(N)$ is the sum of the first N squares. Last night, you dreamed that you derived a “nice” formula for $S(N)$. However, when you woke up in the morning, you forgot the formula. Nevertheless, you clearly remember that $S(N)$ is a polynomial in N and has degree exactly three. Write $S(N)$ as a linear combination of polynomials in $\mathcal{B}$.

$$\text{Want: } \lambda_i \text{ s.t. } \lambda_0 q_0(N) + \lambda_1 q_1(N) + \lambda_2 q_2(N) + \lambda_3 q_3(N) = S(N)$$

$$\lambda_i = \langle S(N), q_i(N) \rangle = S(i)$$

Since $\{q_0, q_1, q_2, q_3\}$ is orthonormal,

$$\lambda_0 = S(0) = 0^2 = 0$$

$$\lambda_1 = S(1) = 0^2 + 1^2 = 1$$

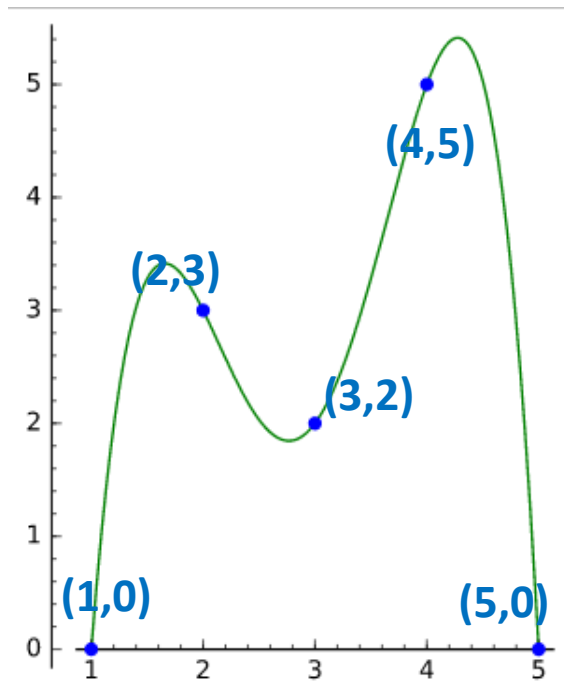
$$\lambda_2 = S(2) = 0^2 + 1^2 + 2^2 = 5$$

$$\lambda_3 = S(3) = 0^2 + 1^2 + 2^2 + 3^2 = 14$$

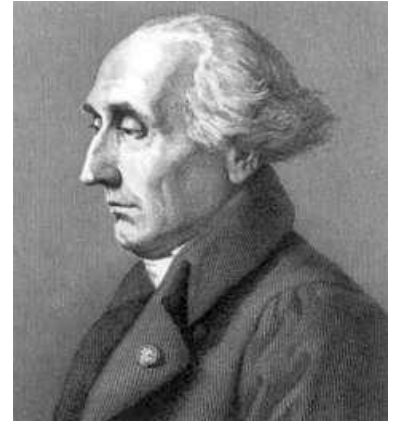
Lagrange Polynomials

Example of Orthogonal Basis

Lagrange Polynomials (1795)



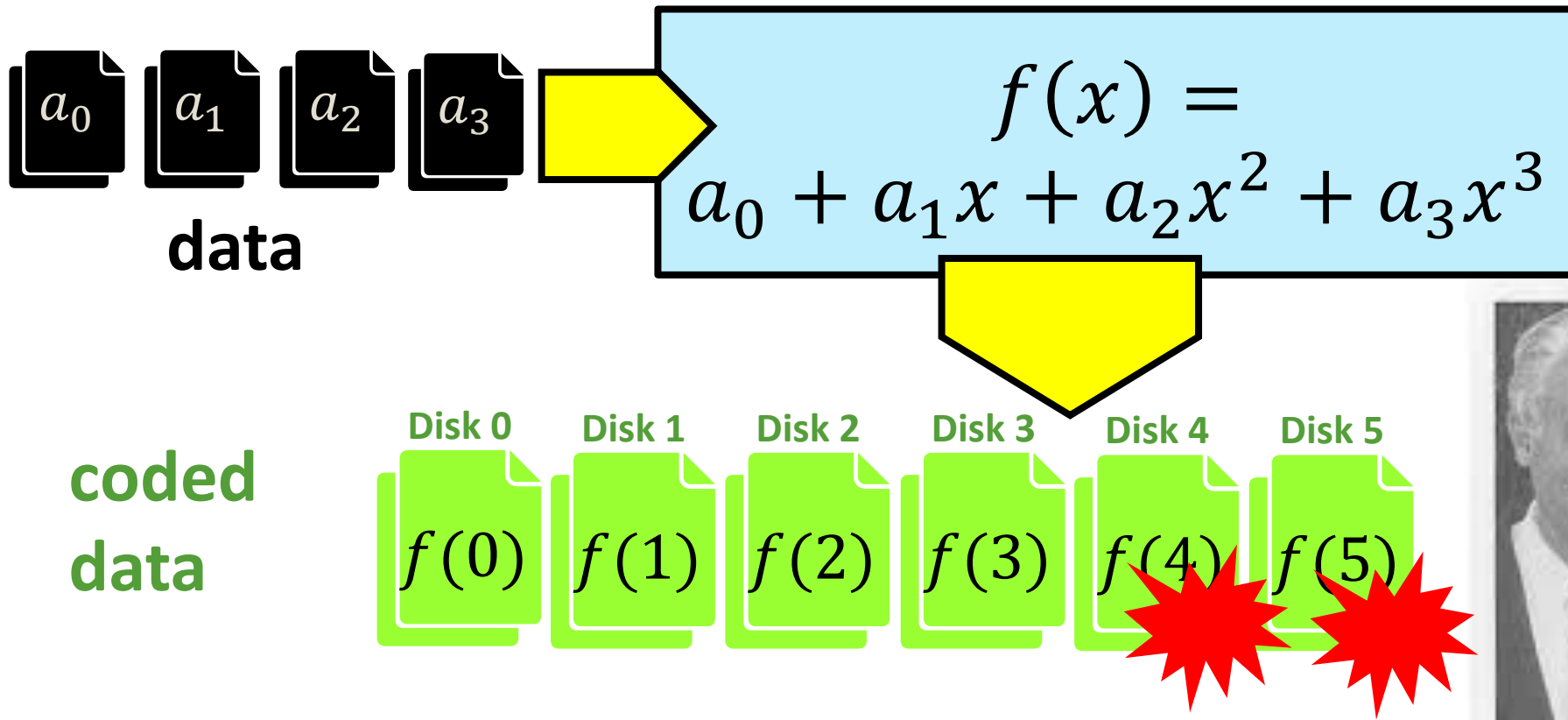
Given $d + 1$ points $(\alpha_1, \beta_1), \dots, (\alpha_{d+1}, \beta_{d+1})$, there is a formula to compute the unique polynomial $f(X)$ of degree at most d such that

$$f(\alpha_i) = \beta_i \text{ for all } 1 \leq i \leq d + 1.$$


Joseph-Louis
Lagrange

$$f(X) = \sum_{i=1}^{d+1} \beta_i \frac{(X - \alpha_1)(X - \alpha_2) \cdots (X - \alpha_{i-1})(X - \alpha_{i+1}) \cdots (X - \alpha_{d+1})}{(\alpha_i - \alpha_1)(\alpha_i - \alpha_2) \cdots (\alpha_i - \alpha_{i-1})(\alpha_i - \alpha_{i+1}) \cdots (\alpha_i - \alpha_{d+1})}$$

Reed-Solomon Codes (1960)



Irving S. Reed and Gustave Solomon

When Disks 4 and 5 fail, we can download $f(0), f(1), f(2), f(3)$ from Disks 0 to 3. Using Lagrange polynomials, we can find $f(4)$ and $f(5)$.

Coded Computation of Multiple Functions

Wilton Kim, Stanislav Kruglik, Han Mao Kiah

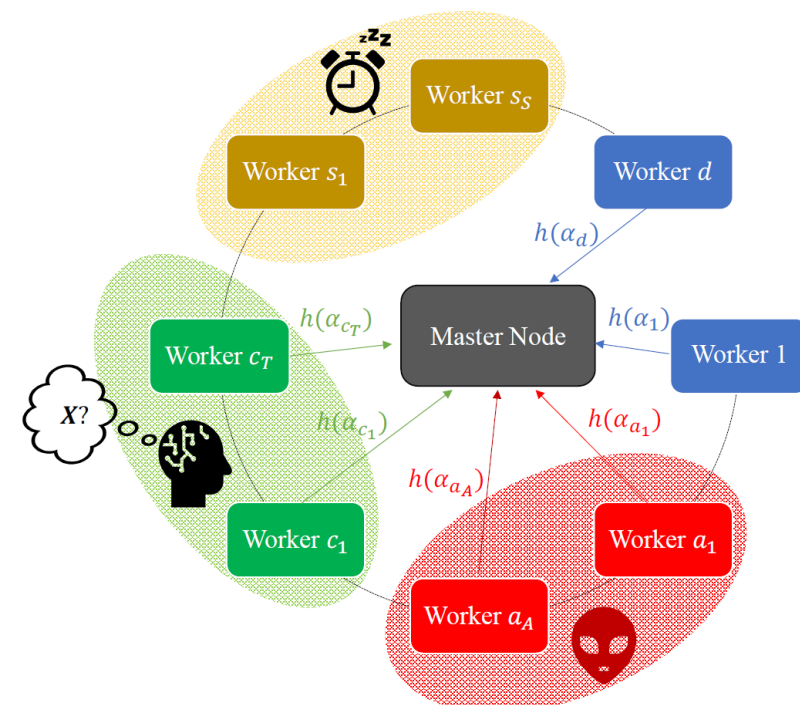
Abstract: We consider the problem of evaluating arbitrary multivariate polynomials over several massive datasets in a **distributed computing** system with a single master node and multiple worker nodes.

We focus on the general case when each multivariate polynomial is evaluated over its dataset and propose a generalization of the **Lagrange Coded Computing** framework (Yu et al. 2019) to provide robustness against stragglers who do not respond in time, adversarial workers who respond with wrong computation and information-theoretic security of dataset against colluding workers. Our scheme introduces a small computation overhead which results in a reduction in download cost and also offers comparable resistance to stragglers over existing solutions.

- <https://arxiv.org/abs/2212.07091>

See also <https://crossminds.ai/video/lagrange-coded-computing-optimal-design-for-resilient-secure-and-private-distributed-learning-60774e5c825a3436b95eee81/>

Video by **Prof. Avestimehr**



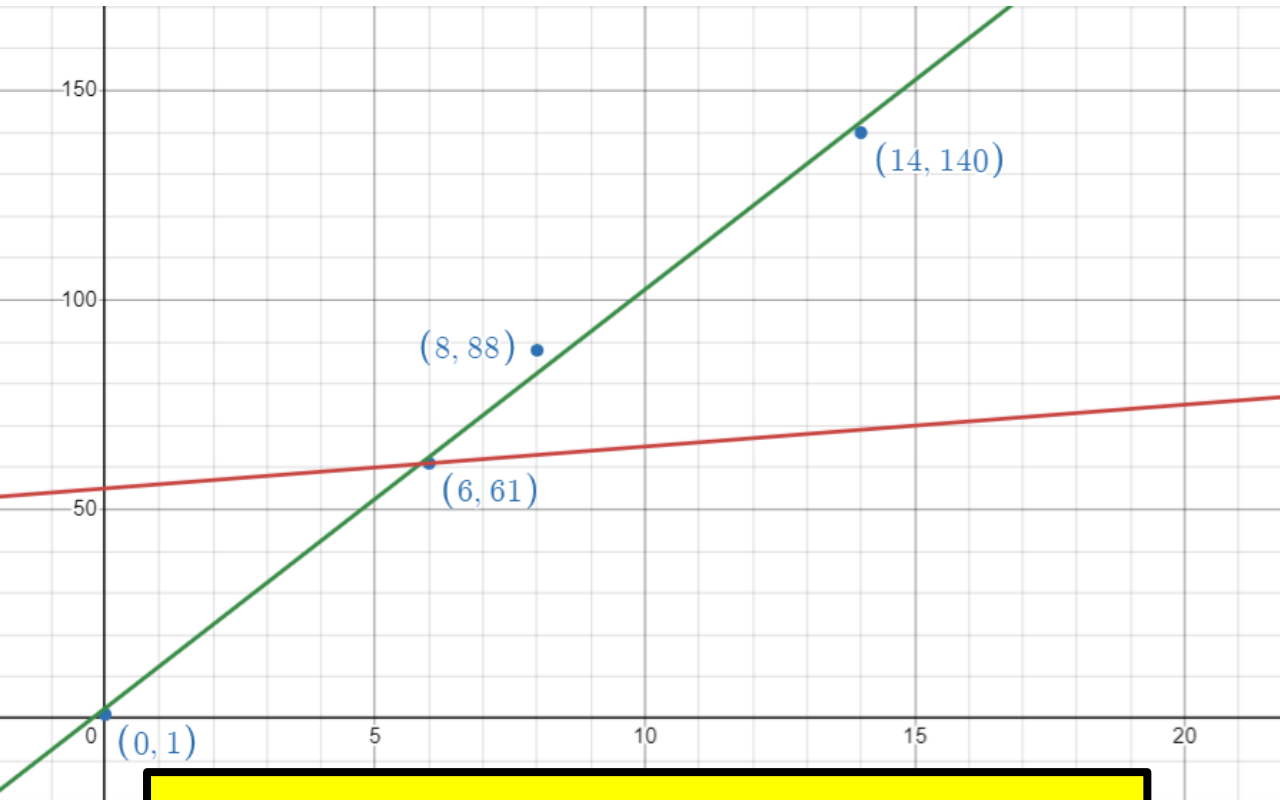
**DISCLAIMER: I am NOT taking
any research students for
URECA, ODYSSEY, or in any form.**

Least Squares Approximation

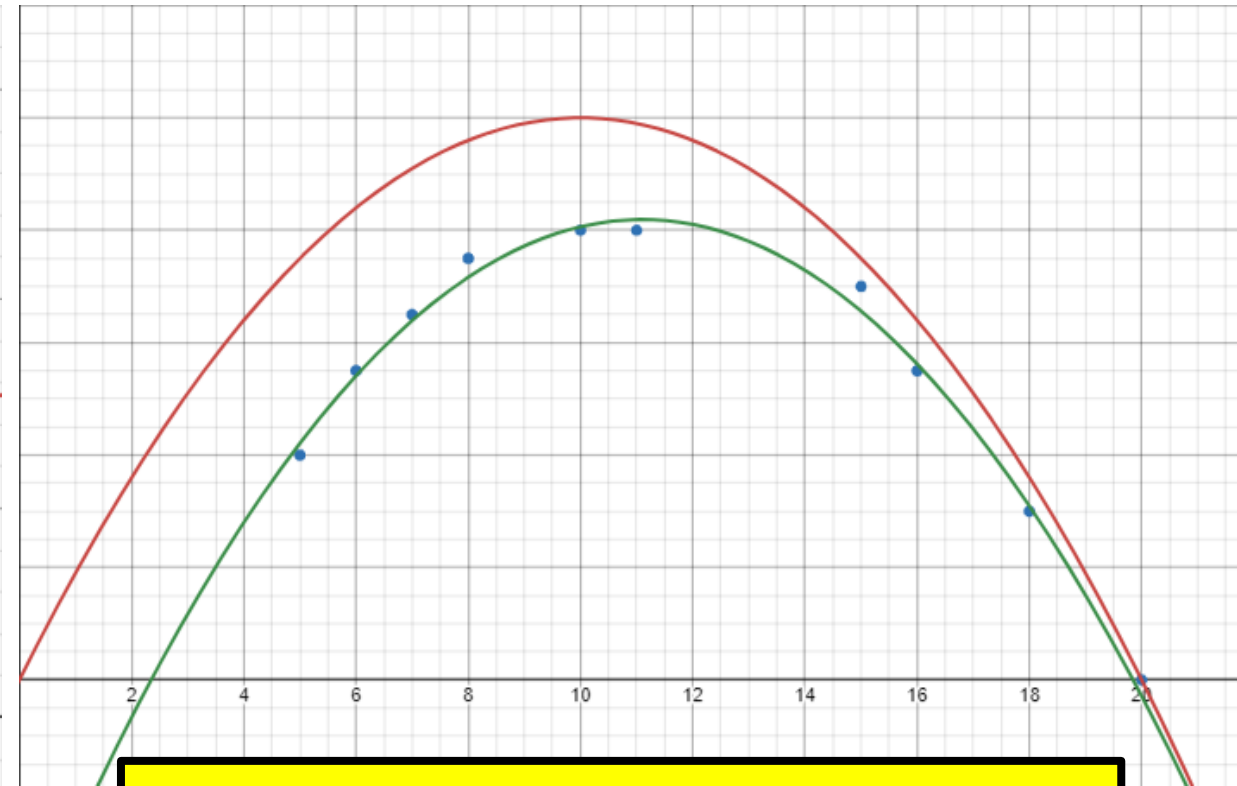
Fourier Epicycles

Another Example of Orthogonal Basis

Least Squares Approximation

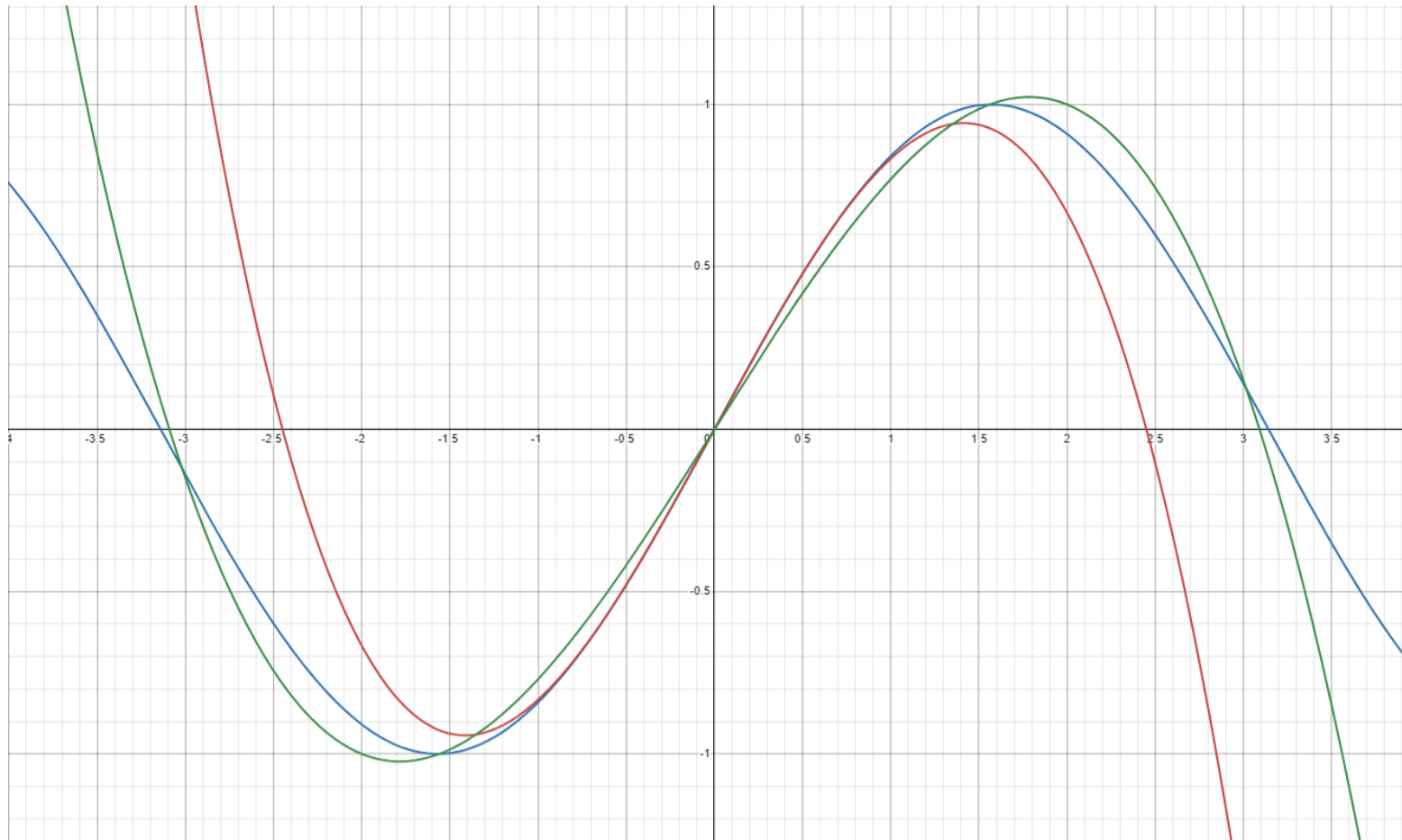


Lec11B: Fitting a straight line.



Tut09 Q8: Fitting a parabola.

Least Squares Approximation of Functions



Let's fit

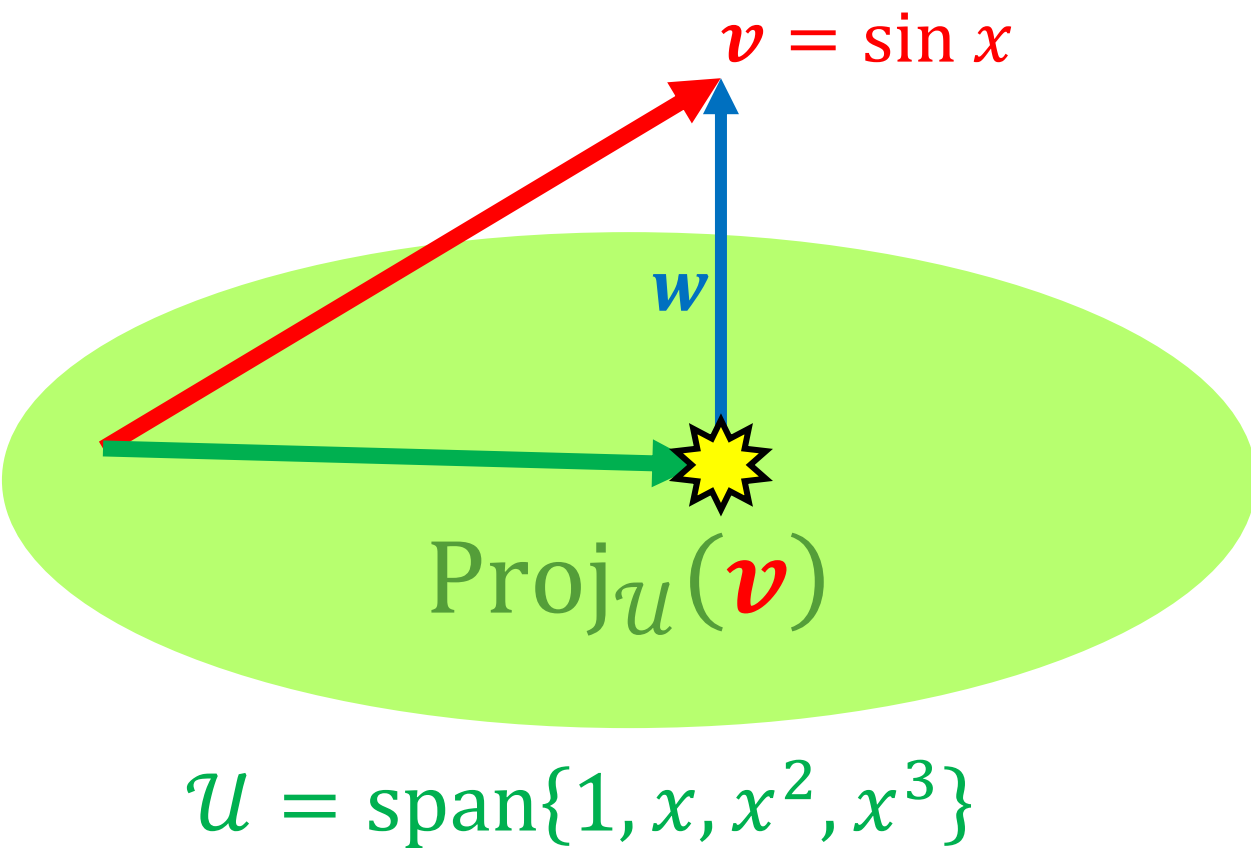
$$f(x) = \sin x$$

using cubic functions...

$$A(x) = x - \frac{1}{6}x^3$$

$$B(x) = 0.86x - 0.09x^3$$

Approximation of Functions



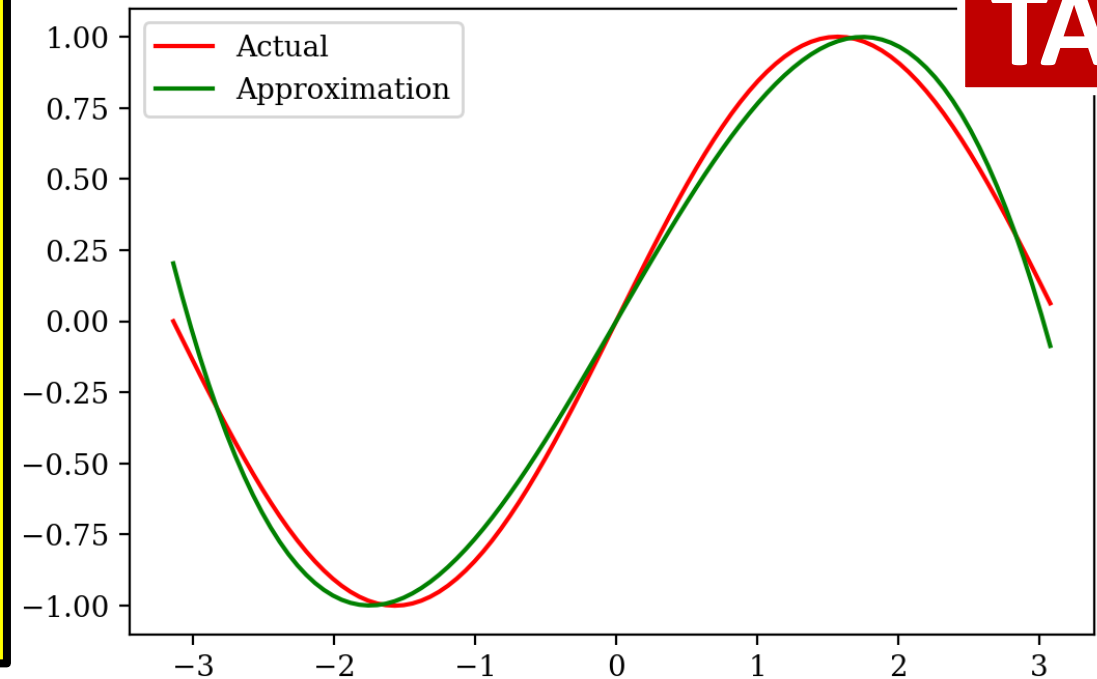
Example.

Consider the inner product $\langle f, g \rangle \triangleq \int_{-\pi}^{\pi} f(x)g(x) dx$ on the space $\mathcal{C}_{\mathbb{R}}[-\pi, \pi]$.

Consider $f(x) = \sin x$ and

the subspace $\mathcal{U} = \text{span}\{1, x, x^2, x^3\}$.

Find $g^* \in \mathcal{U}$ such that $\|f - g^*\| \leq \|f - g\|$ for all $g \in \mathcal{C}_{\mathbb{R}}[-\pi, \pi]$.



An ORTHOGONAL basis is `[1, x, x**2 - 3.29, x**3 - 5.92*x]`

An ORTHONORMAL basis is `[0.40, 0.22*x, 0.14*x**2 - 0.45, 0.09*x**3 - 0.50*x]`

>> Projection on f onto U = `-0.09*x**3 + 0.86*x`

>> ||f - Projection|| = `0.166`

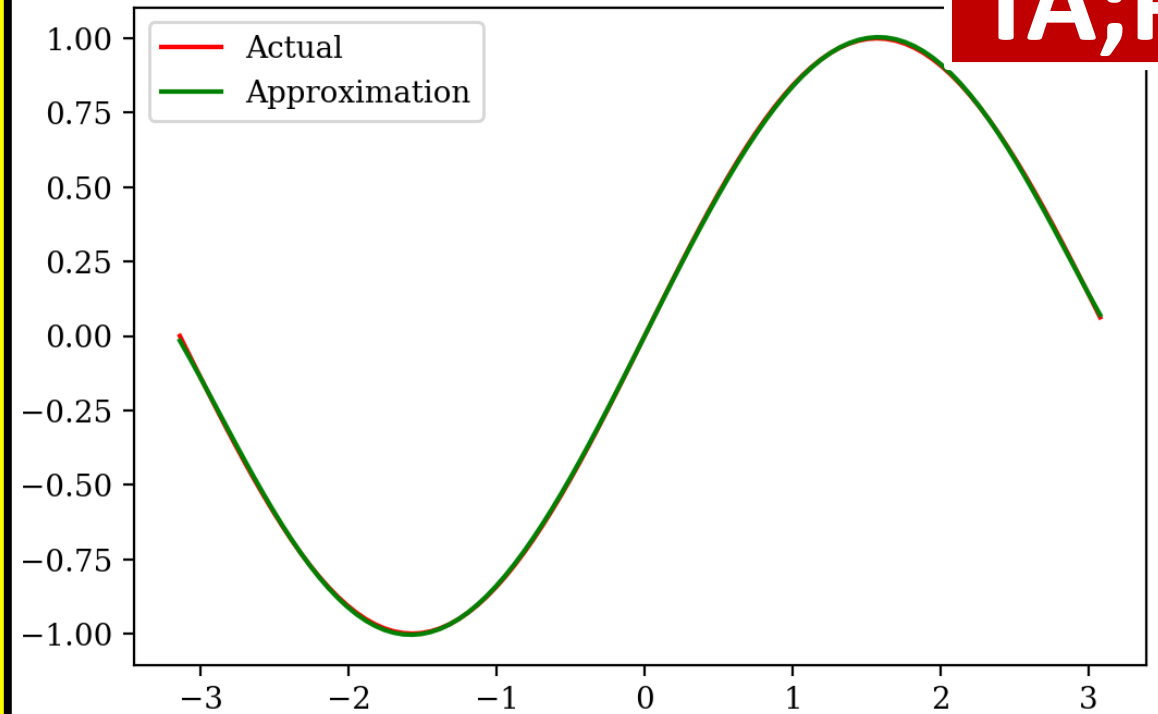
Example.

Consider the inner product $\langle f, g \rangle \triangleq \int_{-\pi}^{\pi} f(x)g(x) dx$ on the space $\mathcal{C}_{\mathbb{R}}[-\pi, \pi]$.

Consider $f(x) = \sin x$ and

the subspace $\mathcal{U} = \text{span}\{1, x, x^2, x^3, x^4, x^5\}$.

Find $g^* \in \mathcal{U}$ such that $\|f - g^*\| \leq \|f - g\|$ for all $g \in \mathcal{C}_{\mathbb{R}}[-\pi, \pi]$.



An ORTHOGONAL basis is `[1, x, x**2 - 3.29, x**3 - 5.92*x, x**4 - 8.46*x**2 + 8.35, x**5 - 10.97*x**3 + 23.19*x]`

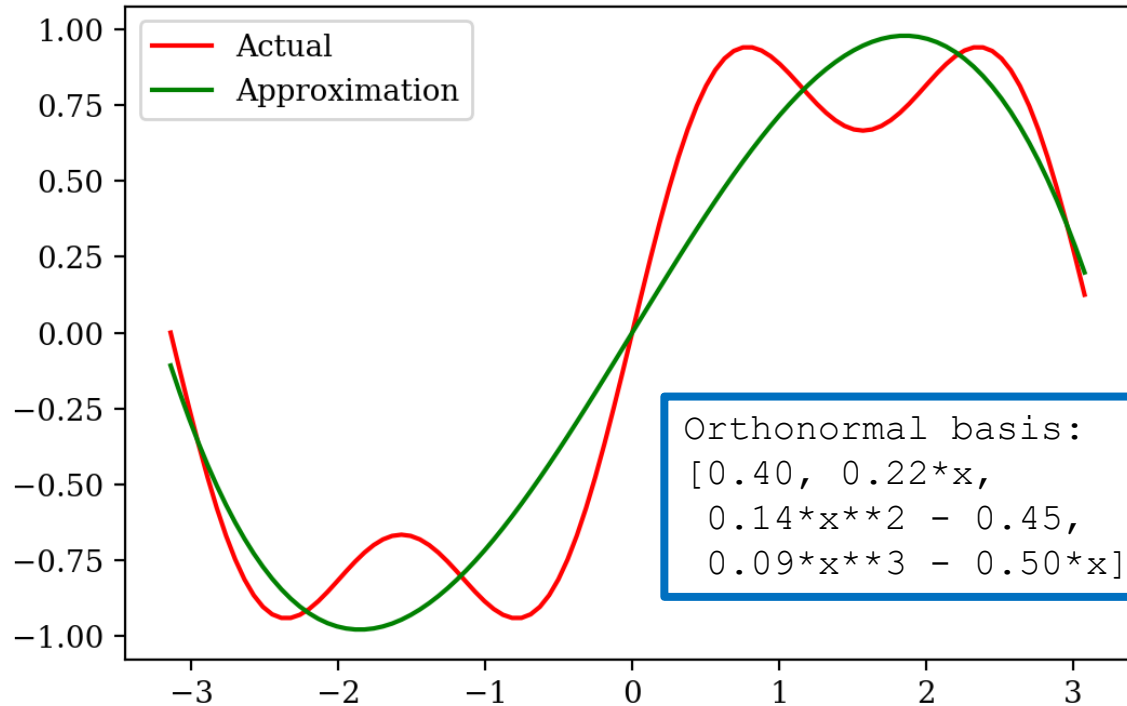
An ORTHONORMAL basis is `[0.40, 0.22*x, 0.14*x**2 - 0.45, 0.09*x**3 - 0.50*x, 0.05*x**4 - 0.45*x**2 + 0.45, 0.03*x**5 - 0.37*x**3 + 0.79*x]`

>> Projection on f onto U = `0.0056*x**5 - 0.16*x**3 + 0.99*x`

>> ||f - Projection|| = `0.0107`

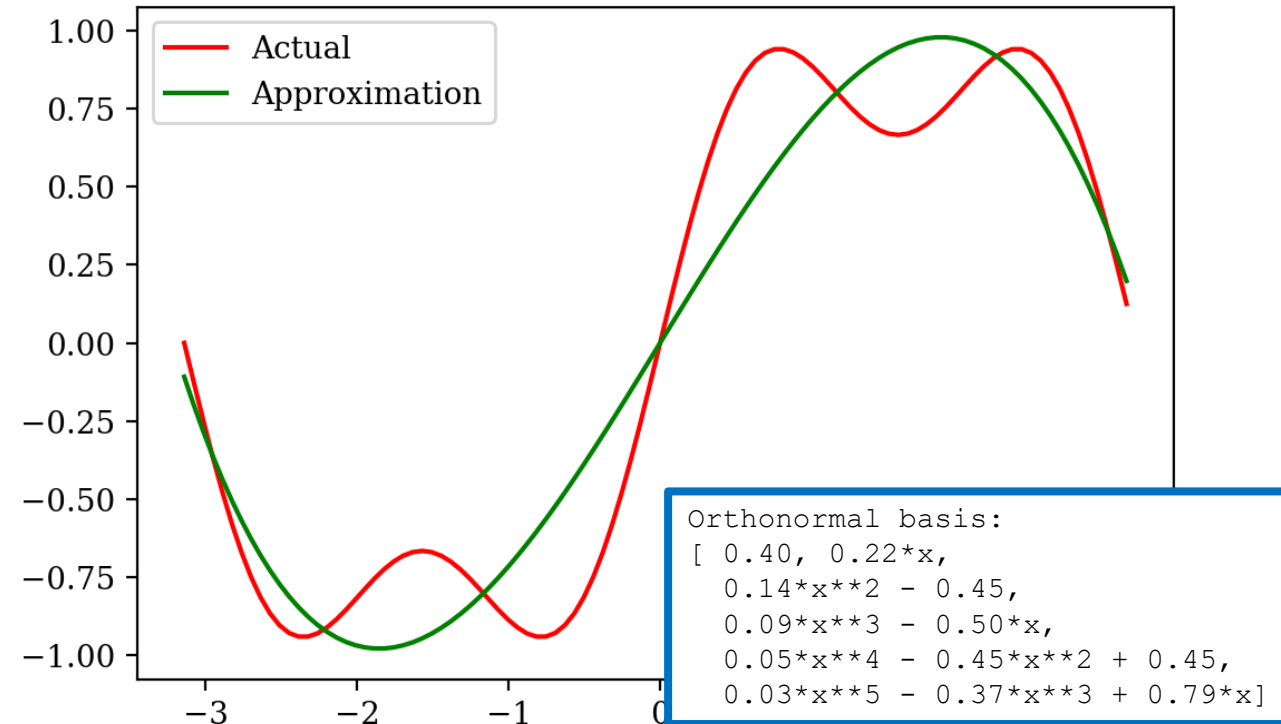
Example. Consider the inner product $\langle f, g \rangle \triangleq \int_{-\pi}^{\pi} f(x)g(x) dx$ on the space $\mathcal{C}_{\mathbb{R}}[-\pi, \pi]$.

Consider some $f(x)$.



When $\mathcal{U} = \text{span}\{1, x, x^2, x^3\}$,
the best approximation has

$$\|f - g^*\| = 0.564.$$



When $\mathcal{U} = \text{span}\{1, x, x^2, x^3, x^4, x^5\}$,
the best approximation has

$$\|f - g^*\| = 0.544.$$

Theorem. Complex Fourier Basis.

Consider $\langle f, g \rangle = \int_0^{2\pi} f(z) \overline{g(z)} dx$ on the space $\mathcal{C}_{\mathbb{C}}[0, 2\pi]$.

For all integers n , set $q_n(z) = \frac{1}{\sqrt{2\pi}} e^{niz}$.

Then $\mathcal{F} = \{\dots, q_{-2}(z), q_{-1}(z), q_0(z), q_1(z), q_2(z), \dots\}$ is orthonormal.

In other words,

$$\langle q_i(z), q_j(z) \rangle = \begin{cases} 0, & \text{if } i \neq j. \\ 1, & \text{if } i = j. \end{cases}$$

Some complex facts.

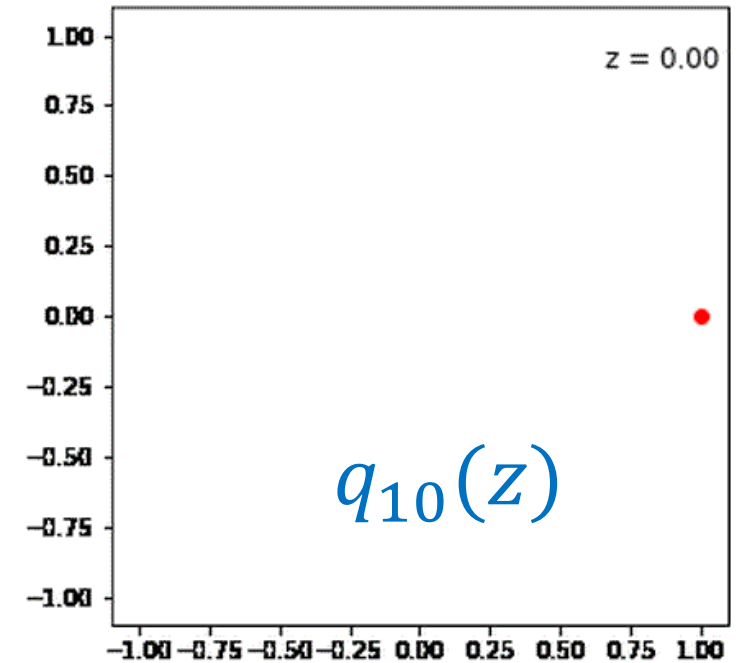
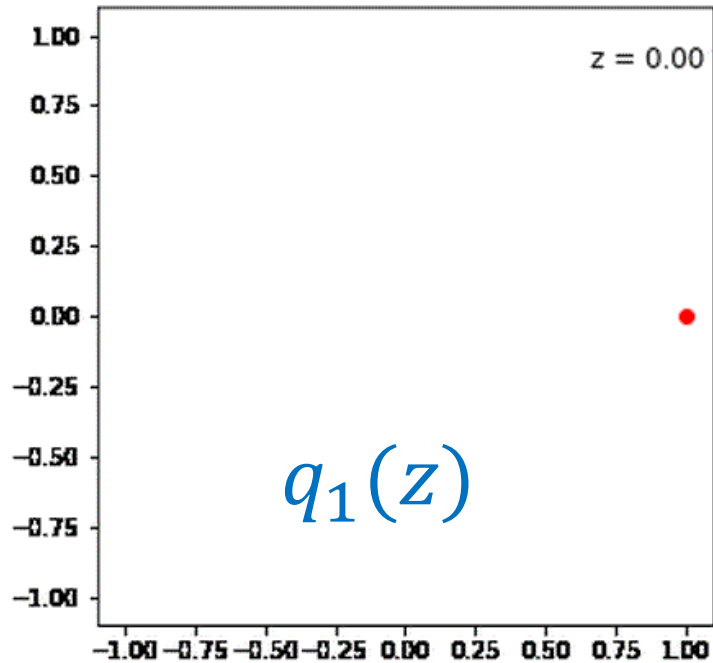
- $\int_0^{2\pi} e^{niz} dz = 0$ if n is a nonzero integer.
- $\int_0^{2\pi} 1 dz = 2\pi$.
- $\overline{e^{iz}} = e^{-iz}$ if z is real.

Theorem. Complex Fourier Basis.

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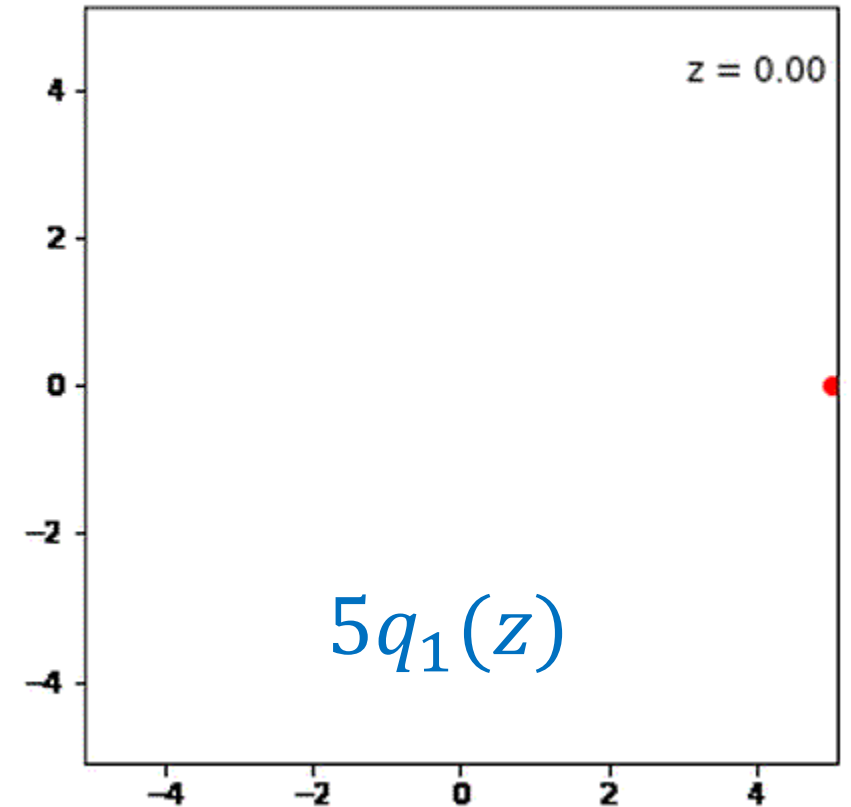
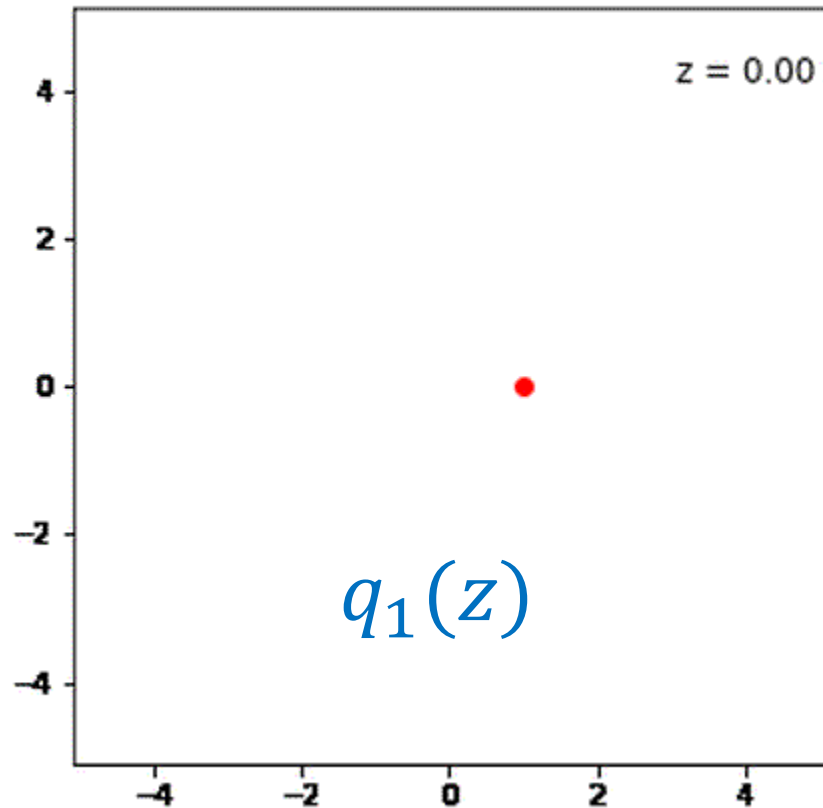


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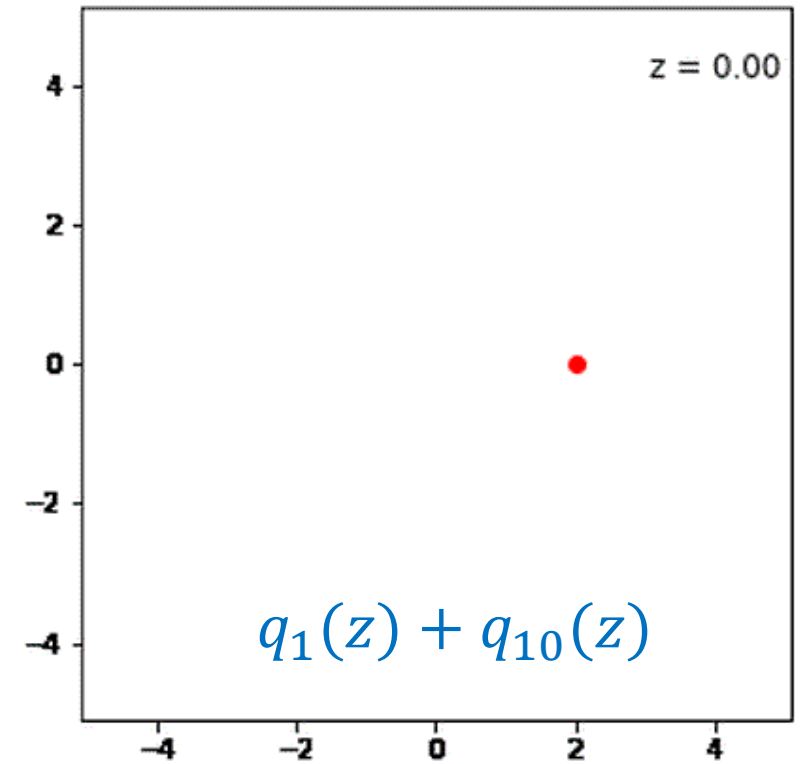
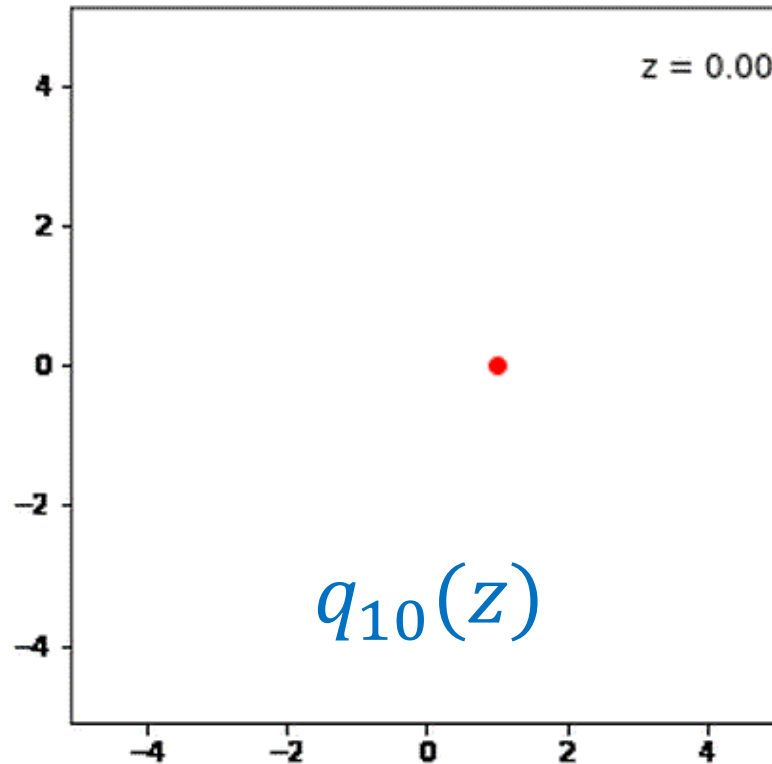
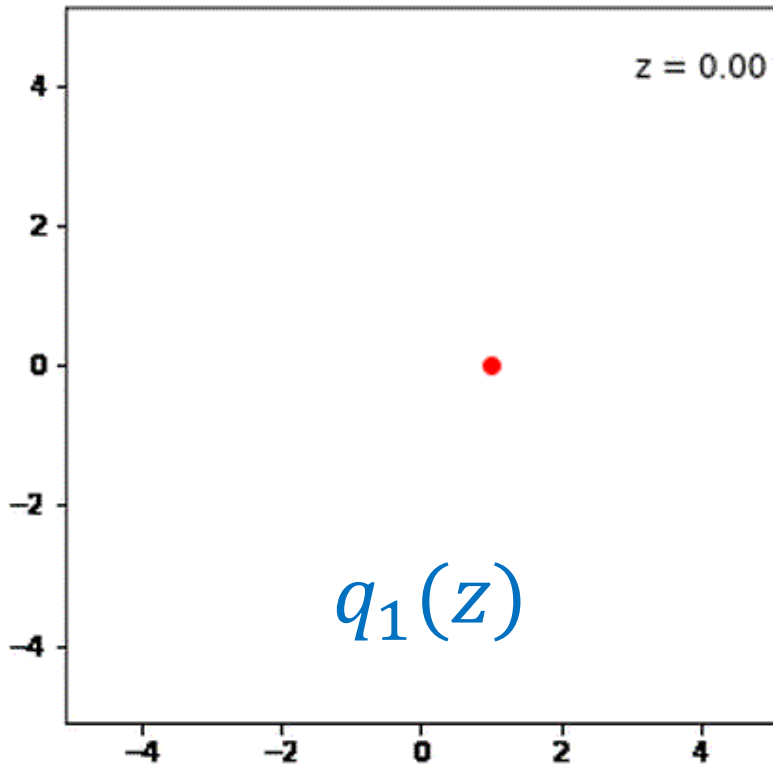


Theorem. Complex Fourier Basis.

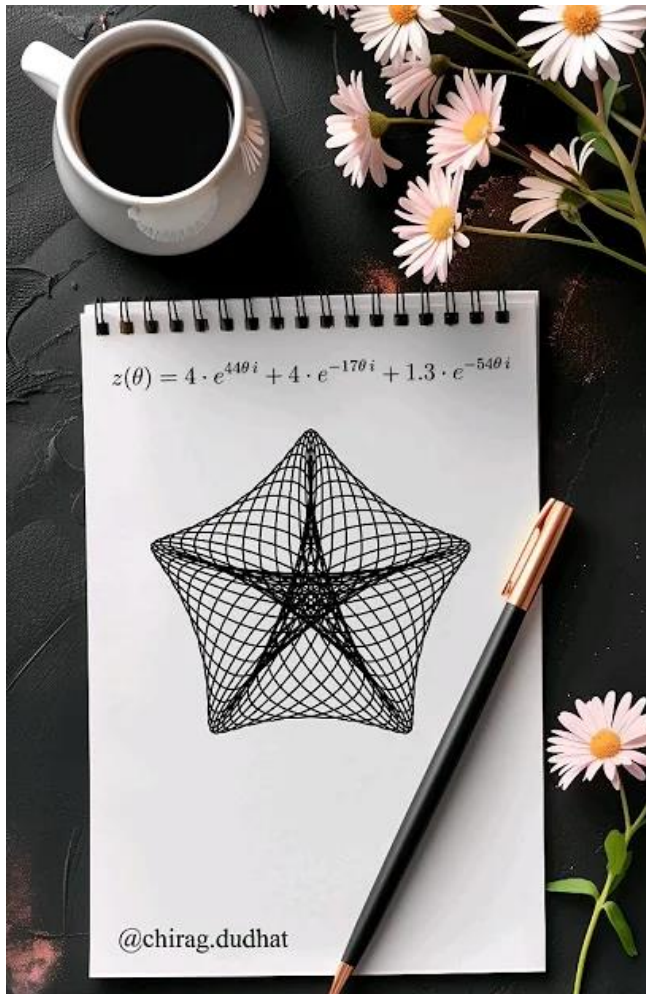
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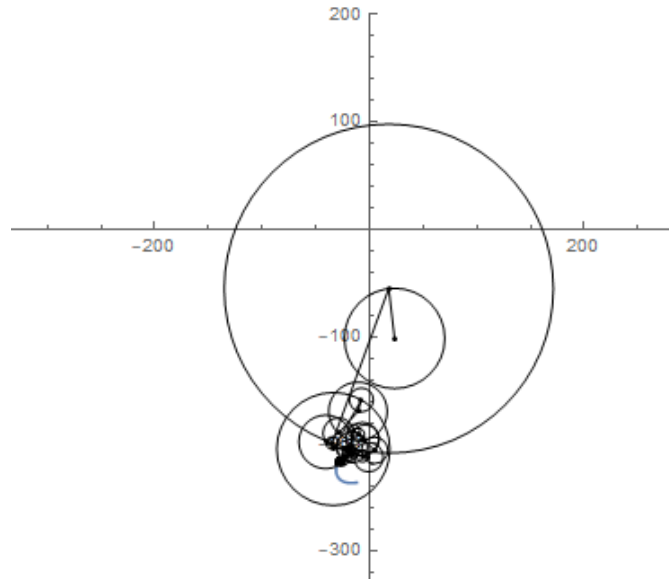
Then $\mathcal{F} = \{\dots, q_{-2}(z), q_{-1}(z), q_0(z), q_1(z), q_2(z), \dots\}$ is orthonormal.



Span of Complex Fourier Basis = Drawing with Circles



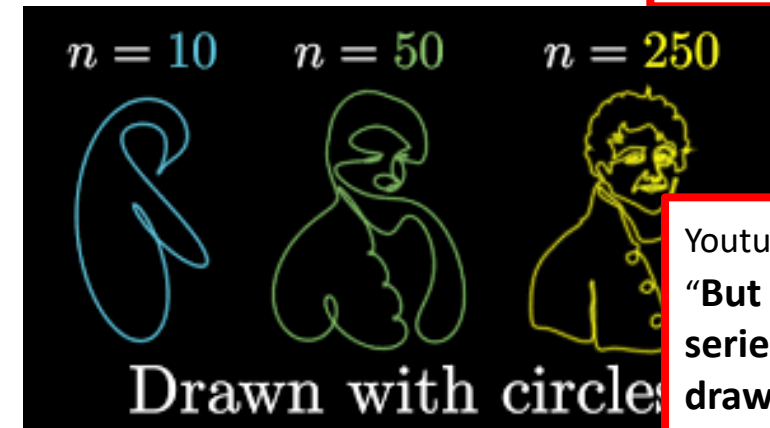
Youtuber: Chirag Dudhat
@fascinating.fractals



<https://mathematica.stackexchange.com/questions/171755/how-can-i-draw-a-homer-with-epicycloids>



Youtuber: Mathologer
"Epicycles, complex Fourier series and Homer Simpson's orbit"



Youtuber: 3Blue1Brown
"But what is a Fourier series? From heat flow to drawing with circles | DE4"

Span of Complex Fourier Basis = Drawing with Circles

Recall: Lec01A

- Complex Fourier Basis / Epicycles
<https://www.desmos.com/calculator/8g6wkimqun>
- Can we get all closed loops using linear combinations?
- Given a closed loop, what is the linear combination?

$$x_7 = \cos(7t)$$

$$x_8 = \cos(8t)$$

$$x_9 = \cos(9t)$$

$$x_{10} = \cos(10t)$$

$$y_3 = \sin(3t)$$

$$y_4 = \sin(4t)$$

$$y_5 = \sin(5t)$$

$$y_6 = \sin(6t)$$

$$y_7 = \sin(7t)$$

$$y_8 = \sin(8t)$$

$$y_9 = \sin(9t)$$

$$y_{10} = \sin(10t)$$

$$u_1 = \cos(-t)$$

$$u_2 = \cos(-2t)$$


 (X_1, Y_1)
 $0 \leq t \leq 2\pi$

47

$$X_1 = 2v_2 - 3v_1 + 18y_1 + y_2 - y_3 + 2y_4 - y_5 + y_6$$

48

$$Y_1 = -2u_2 + 3u_1 - 18x_1 - x_2 + x_3 - 2x_4 + x_5 - x_6$$

